

15) GPE is linearly dependent on height, so it is a straight line. KE is max at the position $x=0$ and zero at $x=$ max displacement. At any point, the $KE+EPE+GPE =$ total energy (constant).
 Answer D

16) $v_{max} = \omega x = (2\pi f) x = 4.71 \text{ ms}^{-1}$